

Chapter 41

Performance Analysis of SDN-Based Intrusion Detection Model with Feature Selection Approach



Samrat Kumar Dey, Md. Raihan Uddin and Md. Mahbubur Rahman

1 Introduction

Today's more advanced and more dynamic applications can no longer be handled effectively by traditional network architecture. Traditional network architecture has been in use for the past decades. But, it is becoming less effective gradually for modern-day applications. To deal with the inadequacy of traditional network architecture, a dynamic and scalable networking architecture, namely Software-Defined Networking architecture (SDN) [1] has emerged. SDN shifts from the fully distributed conventional networking model to a more centralized one. Separation of the data and control planes is a defining characteristic of SDN. Forwarding components namely switches, routers, etc., are elements of the data plane and the controller is an element of the control plane. SDN has been gaining attention recently for both academia and researchers. But, the concept of SDN is not something new. In fact, from the mid-1990s, it started to evolve. A concrete implementation of SDN has been possible thanks to OpenFlow [2] protocol. From the beginning, OpenFlow has been the leading standardized interface between SDN controllers and switches or other data paths [3]. OpenFlow protocol standardizes traffic management and specifies a method for controller communication with network devices. In SDN, network policies are determined by the centralized controller where all the control operations take place. Real-time network information retrieval is straightforward using OpenFlow protocol due to the flow-based forwarding model of SDN. The blessings of SDN have

S. K. Dey (✉) · Md. Mahbubur Rahman
Military Institute of Science and Technology, Dhaka 1216, Bangladesh
e-mail: samratcsepstu@gmail.com

Md. Mahbubur Rahman
e-mail: mahbubcse@yahoo.com

Md. Raihan Uddin
Daffodil International University, Dhaka 1207, Bangladesh
e-mail: mkraihan13@gmail.com

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already been demonstrated by SDN backbones like Google B4 [4] and Huawei carrier network [5]. NOX [6], Ryu [7], Beacon [8], Open Daylight [9], and Floodlight [10] are some examples of SDN controller software that open-source communities have already devised. However, with these major advances, SDN also brings in various security issues related to the application interface and control data. Consequently, security has become an important point of concern for SDN, hence, deserve significant consideration from industry and academia. Intrusion detection guards a network against being affected by malicious pieces of software. There are, conventionally, two kinds of intrusion detection scheme, namely signature-based detection and anomaly-based detection. In signature-based detection, new data is compared with a database of known and discovered intrusions. In contrast, in anomaly-based detection, new data is compared against a model of normal activity and flagged as anomaly only if the data diverges considerably from the model. Consequently, new and unknown intrusion can be detected successfully in anomaly-based approach. In many areas of computer science such as image processing, speech recognition, and face detection, machine learning is being used successfully. Intrusion detection approaches leveraging machine learning are gaining success gradually. These approaches can find interconnection in data automatically and therefore, can be used to build powerful intrusion detection system. In this exploration, we have analyzed the standard NSL-KDD dataset for predicting possible intrusion by using some machine learning algorithms. Most effective classification methods, namely J48, Random Forest, Projective Adaptive Resonance Theory (PART), Naive Bayes, Decision Table (DT), Radial Basis Function Network (RBFN), and Bayesian Network have been applied. Finally, for the performance improvement of the classifier, utmost prevalent feature selection methods are applied as data preprocessing.

2 Related Work

Nowadays, detection of flow-based anomaly is researched comprehensively. This section presents various review of recent accomplishment related to SDNs and Machine Learning (ML). In these days, Flow-based anomaly detection systems have been extensively researched. In [11], a flow-based anomaly detection system was proposed, which is based on Multilayer Perceptron (MLP) neural network with one hidden layer and Gravitational Search Algorithm (GSA). This model can differentiate between normal and malicious flows with reasonable accuracy. In [12], the authors introduce an inductive NIDS using One-Class Support Vector Machines for analysis. Unlike other systems, their model is trained with malicious network data, thus, achieved a low false alarm rate. The authors of [13] implemented four leading traffic anomaly detection algorithms (threshold random walk with credit-based rate limiting, rate limiting, maximum entropy detector, and NETAD) in an SDN environment using NOX controller and OpenFlow compliant switches. But, according to the results of their experiments, it is evident that these algorithms are more successful in identifying malicious activities in the Small Office or Home Office (SOHO)

networks but not in the Internet Service Provider (ISP). A traffic flow features based and lightweight DDoS attack detection system is demonstrated in [14]. Leveraging programmatic interface of NOX platform traffic flow information is obtained with a very low overhead. Using Self-Organizing Maps, this method detects DDoS attacks at a high rate. Kokila RT et al. [15] also devised method for detecting DDoS attack analyzing the SVM classifier. According to their experiments, this method produces less false positive rate in comparison with other techniques. An optimized protection mechanism (OpenFlowSIA) using SVM classifier is proposed by Trung et al. [16]. Along with SVM classifier, this mechanism uses Idle-timeout Adjustment (IA) algorithm devised the same authors and can detect DDoS attack in SDN. Tang et al. [17] present a Deep Neural Network-based intrusion detection systems with only six features and achieve a high degree of accuracy.

3 Methodology

3.1 Proposed ML-Based Classification Model

Our aim is to build a feasible detection scheme which can produce small false alarm rate with a very high accuracy. Here, we present our proposed model that establishes an effective intrusion detection scheme for classifying intrusion data. This intrusion detection scheme provides high detection rate and low false alarm rate. A hybrid classification model based on two layers is shown in Fig. 1. The first layer which is based on some common feature selection methods eliminates unrelated and inconsequential features and provides the selected features to the second layer. The second layer, then, classifies the abridged data set using some useful machine learning algorithm. The model is trained using 10 fold cross-validation technique. Assessment of the designed scheme carried out by using some appropriate measures.

3.2 Machine Learning Approach

According to [18], the common idea behind most machine learning is that a system learns to accomplish a task by studying a training set of examples. The system of distributed computers and controllers then performs a similar task with data that has not encountered before. The main idea of using ML in flow-based anomaly detection is to build a predictive model automatically based on the training dataset. Machine learning can be enumerated as the study of algorithms that advance their enactment with experience and the machine takes every required step accomplished likewise in a mentioned way. Currently, ML algorithms have been extensively used for a number of classification and prediction problems. The below Fig. 2 shows complete anomaly detection flowchart mechanism in OpenFlow controller.

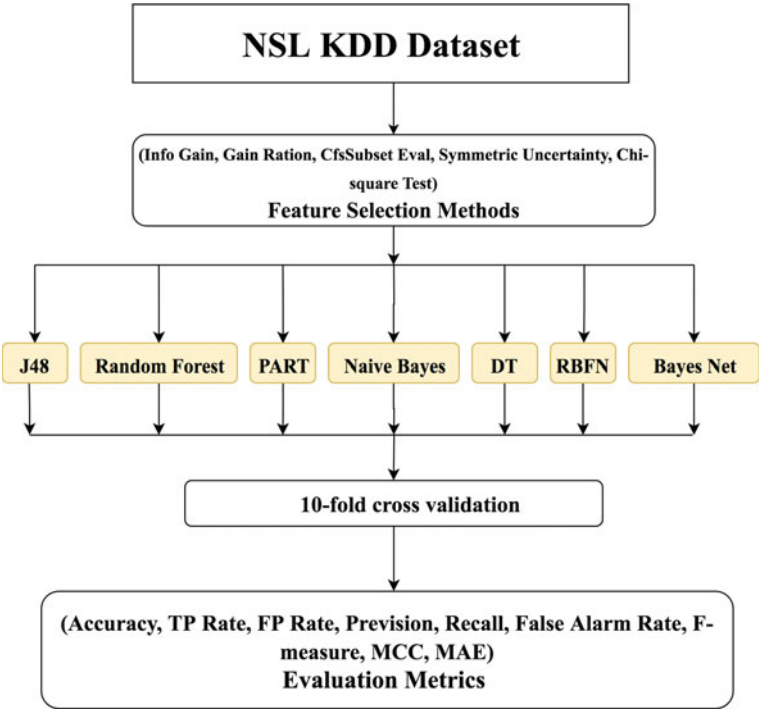
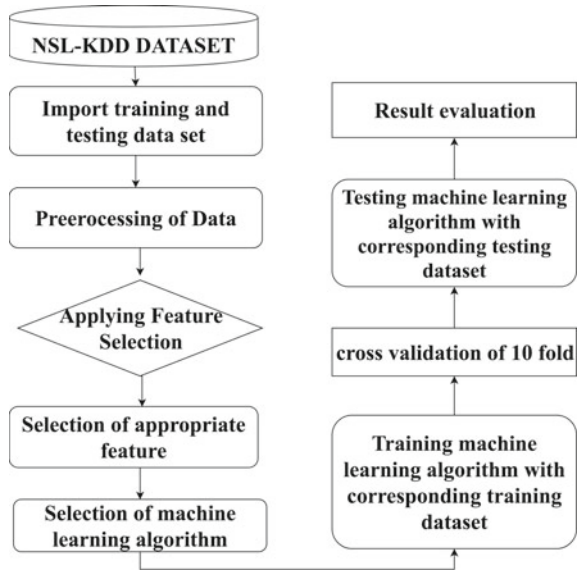


Fig. 1 Proposed machine learning-based classification Model

Fig. 2 Anomaly detection flowchart using machine learning algorithms



3.3 *NSL-KDD Dataset*

Choosing an accurate data set for establishing and evaluating a model is by no means straightforward. Our intent is to select an extensive, free of noise, consistent, and redundancy-free data set. For different anomaly detection systems, various types of dataset have been used; some are self-made while others are publicly accessible. KDDCup-99 is publicly accessible among others which is most extensively used and accepted dataset. KDDCup-99 is practically free for download, but the whole dataset is so big that it dramatically increases the cost of computation of the Intrusion Detection. Further, the authors in [19] claim that only 10% of the dataset is usually used. In training dataset, there are many extraneous data against which there are identical records in the testing dataset. Consequently, the process of learning of the system is sometimes distracted. NSL-KDD is a refined version of KDDCup-99, which minimizes the redundancies between training dataset and testing dataset. Tavallace et al. [20] recommended NSL-KDD dataset which consists of intrusion data. Many researchers are using the dataset for their experiment. There are 41 features in NSL-KDD that contains both normal and attack patterns. There are 5 normal classes and 4 attack classes. These 4 attack classes are Denial of Service (DoS), User to Root (U2R), Probe, and Remote to Local (R2L). KDDTrain Dataset contains 125,973 network traffic samples and 22,554 traffic sample is contained by KDDTest Dataset.

3.4 *Feature Selection*

Accuracy of anomaly detection decreases in the presence of redundant attributes in the intrusion dataset. Therefore, developing methods for suitable feature selection that can sort out unrelated attributes is a research challenge. Feature selection is the technique of choosing relevant features and removing irrelevant features from the dataset to accomplish a certain task. Decreasing the number of redundant, irrelevant, and noisy features can lead to a better model by speeding up a data mining algorithm with an improvement in learning accuracy. We refined dataset features using Info Gain, Gain Ratio, CFS Subset Evaluation, Symmetric Uncertainty, and Chi-Square Test. Attribute selection procedures with different evaluators and their search strategies are shown in Table 1.

3.5 *Evaluation Metrics*

Performance of intrusion detection rate is measured in terms of accuracy (AC), precision (P), recall (R), F-measure (F), False Alarm Rate (FAR), and Mathews correlation coefficient (MCC). To calculate the value of these techniques, some performance metrics derived from confusion matrix are taken into account.

Table 1 Feature selection with different evaluator and search method

Evaluator	Search	Selected attributes
Info gain	Ranker	5, 6, 3, 4, 33, 35, 34, 40, 41, 23, 30, 29, 12, 27, 28:15
Gain ratio	Ranker	28, 12, 41, 27, 4, 6, 5, 30, 29, 40, 3, 25, 26, 39, 34:15
CFS subset evaluator	Best first	5, 6, 12, 25, 28, 30, 31, 37, 41:9
Symmetric uncertainty	Ranker	6, 5, 4, 41, 28, 12, 27, 30, 3, 40, 29, 34, 35, 33, 37:15
Chi-square test	Ranker	5, 6, 3, 33, 35, 34, 4, 40, 23, 12, 41, 30, 29, 27, 37:15

According to confusion matrix, True positive is the total number of samples predicted as normal while they were actually normal and false negative is the total number of samples predicted as normal while they were actually attacked. On the other hand, false positive is the opposite of true positive and it is the total number of samples predicted as attack while they were actually normal. True negative is the total number of samples predicted as attack while they were actually attack.

Accuracy (AC): shows the proportion of the classification overall N Examples that were correct.

$$AC = \frac{TP + TN}{TP + FP + TN + FN}$$

Precision (P): shows the proportion of the network intrusion detection system detected intrusion that are real intrusion. The higher the value of P, the lower the false alarm rate.

$$P = \frac{TP}{TP + FP}$$

Recall (R): shows the proportion of correctly classified positive examples. We are in search of a high value of R.

$$R = \frac{TP}{TP + FN}$$

F-measure (F): By conveying balance between accuracy and recall, it gives a better measure of accuracy. We are in quest of a high F-measure value.

$$F = \frac{2}{\frac{1}{P} + \frac{1}{R}}$$

Mathew's correlation coefficient (MCC): It returns a binary value between -1 and 1 by showing the value correlation coefficient between the predicted and observed binary classifications.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

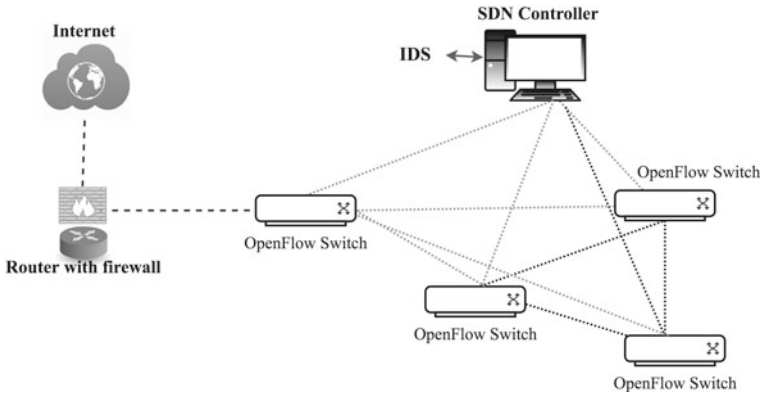


Fig. 3 Proposed flow-based anomaly detection architecture in SDN

3.6 SDN-Based Flow-Based Anomaly Detection Architecture

In general, OpenFlow switches are monitored by SDN controller. SDN controller can request for all network data whenever necessary. Therefore, we implemented our proposed intrusion detection segment in SDN controller, which is illustrated in Fig. 3. When network data is required, an OpenFlow stats request message is sent from the controller to all OpenFlow switches. After receiving the request, OpenFlow switch sends an OpenFlow stats reply message with all available data back to the controller. In this architecture, OpenFlow switch handles the incoming packet and makes responses according to the availability of data in flow table using OpenFlow protocol. Once a network anomaly is discovered and recognized, OF protocol can effectively eliminate an intrusion via flow table adjustment.

4 Experimental Results and Analysis

We used WEKA [21] environment and NSL-KDD Dataset for carrying out the experiments. The environment consists of a machine with 4 GB of memory with a Processor of Intel(R) Core(TM) i5-2410M CPU @ 2.30 GHz, 2301 MHz. For loading and analyzing, the dataset heap size of WEKA was increased. To train and test each selected feature, NSL-KDD dataset was used. Tenfold cross-validation technique was used to perform the experiment. Dividing the training set into 10 subsets, we tested each subset when the model is trained on the other 9 subsets. Each subset is processed only once as test data; therefore, the process repeats 10 times. Experimental results of accuracy, True positive rate, False positive rate, Precision, Recall, False alarm rate, F-measure, Mathews correlation coefficient, and Means absolute error of different classifier were achieved. Furthermore, the results of highest accuracy of different

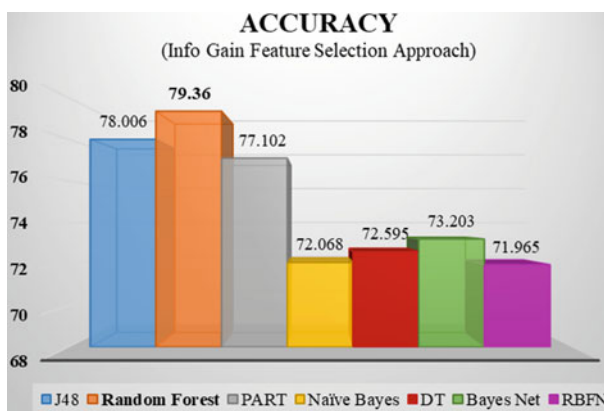


Fig. 4 Highest accuracy is shown by Random Forest with Info Gain feature selection method

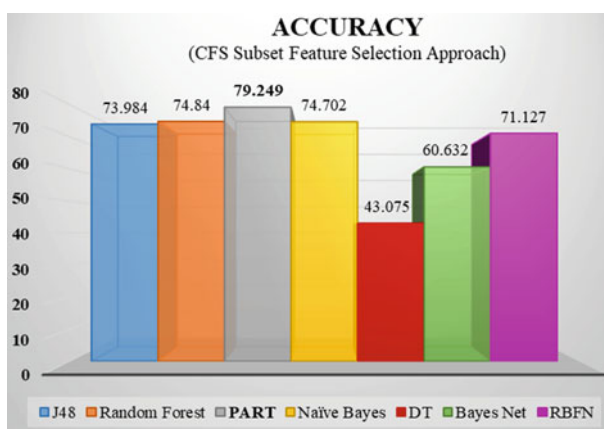


Fig. 5 PART shows the highest accuracy with CFS subset Evaluator feature selection method

classifier in terms of feature selection are shown here. Our aim was to achieve high accuracy with a low false alarm rate. In our experiment, we are capable of successfully attain that. The following Fig. 4 shows the accuracy values of different ML algorithm with Info Gain Feature selection approach where Random Forest classifier produce higher values than other classifier.

As the feature selection search technique changes from Ranker method to Best First, the CFS Subset Evaluator provides the highest accuracy on the PART Classifier with an accuracy of 79.256%, which is represented in Fig. 5.

Figures 6, 7 and 8 shows the highest accuracy achieved by random forest classifier with the feature section technique of Gain Ratio, Symmetric Uncertainty, and Chi-square Test, respectively.

Primarily, we employed the network intrusion detection for two-class-based classification namely normal and anomaly. In addition, we assessed our work by

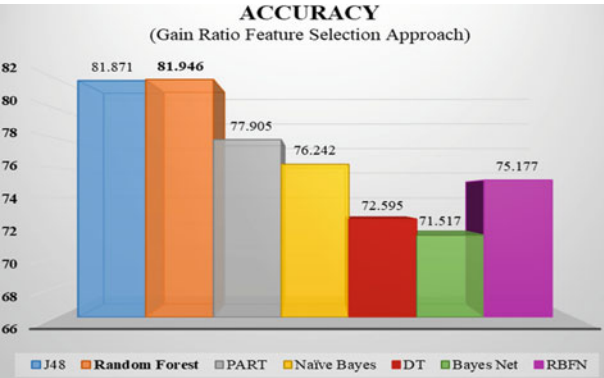


Fig. 6 Highest accuracy is shown by Random Forest with Gain Ratio feature selection method

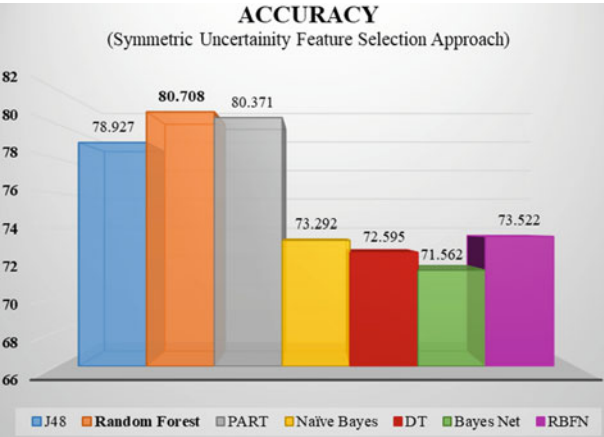


Fig. 7 Highest accuracy is shown by Random Forest with Symmetric uncertainty feature selection method

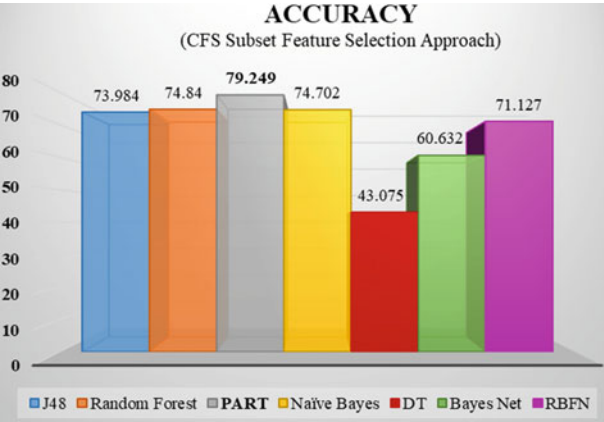


Fig. 8 Highest accuracy is shown by Random Forest with Chi-square test

assimilating the results and proposed a security architecture for detecting flow-based anomaly in OpenFlow Controller.

From the above discussion and also from Fig. 6, it clearly shows that with the feature selection methods of Gain Ratio, Random Forest classifier provides the highest Accuracy of 81.95% with a very low false alarm rate of 0.287%.

Moreover, our results are generated based on the selection of features from a complete dataset of NSL-KDD. Very few approaches have been presented by different authors in order to show the accuracy of machine learning algorithm for NSL-KDD Dataset. For comparing with others we will only consider that approaches which are only based on an NSL-KDD dataset for the detection of a flow-based anomaly in SDN. Authors in [22] shows some approaches for performance analysis of NSL-KDD dataset using Artificial Neural Network. It shows an accuracy of 81.2% for intrusion detection but a suitable way of feature selection is not described clearly. Moreover, no architecture is provided for intrusion detection in SDN. Authors in [23] present a technique of intrusion detection in SDN with an accuracy of 97% using pattern recognition of neural network but the dataset contains only 7 features for testing. For handling intrusion and DDoS attack in SDN, ML-based techniques is presented in [24]. It discussed thoroughly some ML approaches but no performance evaluation using dataset is carried out. An experiment is carried out in [25] using some classification algorithm like J48, SVM, and Naive Bayes showing a high percentage of accuracy but dataset contains only 6 features for testing. Moreover, there also exist some scatter research work on the analysis of NSL-KDD dataset for intrusion detection but no fitting method of feature selection for training and testing has been provided.

5 Conclusion

In this paper, we have presented a machine learning-based classification model for detecting network intrusion and shows the best classifier in terms of different evaluation metrics with different feature selection mechanism. Five feature selection methods with different machine learning algorithm have been analyzed with the support of NSL-KDD dataset. Although our experimental results are not yet high enough comparing with others but it still has significant contribution in the field of appropriate feature selection from a dataset. Our proposed intrusion detection module is capable of easily extract the information about network traffic due to SDN's centralized controller and flexible nature. From the experiment, its clear that Random Forest shows a high test accuracy comparing with all other algorithms therefore for flow-based anomaly detection use of RF is essential in order to achieve high accuracy and speeding up the process of intrusion detection in SDN. In near future, we plan to implement this proposed model in a real Environment of SDN with the real traffic Network.

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